

Privacy-Preserving Few-Shot Traffic Detection Against Advanced Persistent Threats via Federated Meta Learning

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Abstract—Advanced Persistent Threats (APT) utilizes multiple zero-day vulnerabilities to threaten critical industrial infrastructure, having the characteristics of burst, unknown and cross-domain. To resist APT attacks, existing wisdom usually establish a security monitoring platform that remotely links to the cloud-based threat intelligence center. However, the real scenario where few victim users are willing to share raw attack samples considering privacy-preservation, such mentality is hysteretic and cannot identify APT attacks quickly without sacrificing additional incentives. To address this issue, a novel privacy-preserving few-shot traffic detection (PFTD) method based on federated meta learning (FML) is proposed. The PFTD treats the APT detection task as a model generalization optimization process, that transfers the learned knowledge to identify local unknown samples. Client-side models in FML achieve knowledge transferring by two-phase updating over both support dataset and query dataset, while the server-side model obtains global knowledge with model aggregation. These processes compile useful knowledge against APT attacks. With a novel wisdom, we obtained three advantages: 1) High accuracy with a few attack samples; 2) Low latency detection for removing rules matching process; 3) High personalizing to cross-domain APT attacks. Extensive experiments based on multiple benchmark datasets like CICIDS2017 and DAPT 2020 prove the superiority of proposed PFTD.

Index Terms—Advanced persistent threats, federated meta learning, few-shot traffic detection, privacy-preserving.

I. INTRODUCTION

TRAFFIC detection system is an important research field in computer system and network, it is used to detect network intrusion behavior. However, traffic detection systems cannot identify unknown attack behaviors [1]. In recent years, security incidents caused by unknown cyber attacks have emerged

one after another, causing a huge economic stress around the world. On May 2021, Colonial Pipeline, the largest oil product pipeline operator in the United States, was attacked by a ransomware network from the hacker group Darkside. The group exploited the vulnerability of the software to hack into central systems, disrupt and steal the computer data, and also take the pipeline systems that transport oil and gas in major cities on the eastern seaboard offline. Since 2015, OceanLotus, an APT organization in Southeast Asia, has carried out long-term and continuous cyber attacks in China. The organization often exploits 0-day/N-day vulnerabilities during intrusion and lateral movement. During the Russian-Ukrainian war, the cyber-attacks against Ukraine by groups such as APT28 and AgentTesla were strategically used to support ground operations. The spy Trojans and other malicious codes use custom or encrypted protocols to establish covert communication channels and finally bypass the detection of traditional security devices such as firewalls and intrusion detection systems (IDSs). These new types of attacks, especially APT attacks, have brought great challenges to traffic detection technology.

- APT attacks have many characteristics such as burst, unknown and cross-domain. Existing traffic detection methods are limited by the fragmentation of threat intelligence and tend to focus more on discovering locally known attack behaviors.
- Due to privacy concerns or the desire to avoid embarrassment, original APT attack samples cannot be shared timely. As a result, the defenders can only resort to remediation and damage control after the attack has occurred.

To address the above challenges, introducing artificial intelligence (AI), Big Data analysis and cloud technology in the APT detection field has become a trend [2], [3]. For example, AI-based IDS systems aim to extract abnormal/normal features from massive network traffic, and distinguishes whether it is benign or malicious traffic. The Big Data framework enhances the ability of traffic inspection systems in understanding the increasingly complex data distribution of intrusion patterns [4]. The introduction of cloud computing has enhanced the efficiency of traditional traffic detection systems in the face of network attacks with variability, invisibility and unpredictability [5]. Among them, deep learning (DL) is considered to be the most important enabling technique that may have great ability to discover APT attacks. Deep learning-empowered methods utilizes deep neural networks to perform representation learning on network traffic.

Manuscript received 15 January 2023; revised 24 May 2023; accepted 2 July 2023. Date of publication 11 August 2023; date of current version 30 April 2024. This work was supported in part by the National Nature Science Foundation of China under Grants U21B2019, 62202303, and U20B2048, in part by Shanghai Sailing Program under Grant 21YF1421700, and in part by the Defence Industrial Technology Development Program under Grant JCKY2020604B004. Recommended for acceptance by Dr. He Huang. (Corresponding authors : Jun Wu; Gaolei Li.)

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Digital Object Identifier 10.1109/TNSE.2023.3304556

The broad range of applications of deep learning-empowered methods includes both relationship-oriented deep learning in which the detector tends to identify the combination methods of each network attack tools with correlation analysis using Graph Neural Networks [6], [7], and timing-oriented deep learning in which the detector tends to judge the intent of a network attack with the connections between traffic features and time sequence using Recurrent Neural Networks [8], [9].

However, deploying DL-based APT detection systems in cloud-side data centers faces with many challenges. First, cloud-side computing struggles with leakage of the sensitive and confidential information; second, since the cloud-side computing cannot integrate heterogeneous and dispersed threat intelligence due to real-time constraints, it is difficult to discover efficient and real-time unknown attacks. In addition, with the influx of a large number of edge devices, different suppliers require personalized security defence abilities. Therefore, it is particularly important to deploy intelligent APT detection systems to discover unknown attack behaviors at the edge side. Recently, many studies have focused on the construction of edge-side APT detection systems. For example, Yuan et al. [10] designed a CNN-based intrusion detection system and deployed it on the edge devices to identify attack behaviors, which transferred data flows into the form of images. Yet, although the DL-based APT detection technology develops rapidly, there are still numerous challenges when deploying it at the edge side:

- Due to the complexity and dynamic of the Internet environment, it is very difficult for existing traffic detection methods to identify various types of APT attack behaviors according to limited threat intelligence;
- Since massive heterogeneous devices are connected to edge networks, it is difficult for existing APT detection methods to identify the latest evolved attacks;
- Training a chatGPT-like pre-trained model that can fit all local datasets on each edge device is very difficult for the number of APT attack samples is usually nonpublic.

Federated meta-learning (FML) has been proved that it can improve the personalization and generalization capabilities of deep neural networks at edge devices. FML refers to combining many edge devices (meta-learners) to train a large global meta-learner by aggregating model parameters collected on the server. This global learner has a strong generalization ability because it has learnt many samples, a personalized model for each node can be obtained by performing a few steps of gradient descent using the Model-Agnostic Meta-Learning (MAML) algorithm on edge devices. MAML retains the information of different sub-tasks in the form of parameter updates during the training process of each sub-task, so the model trained by meta-learning has better generalization ability [11]. At the same time, FML protects the privacy of each edge device. With the assistance of FML, the global model can be personalized to fit the local data.

Therefore, in order to solve the problem of intrusion detection in APT attacks which is difficult for edge-side smart devices to deal with unknown behavior attacks in the context of federated learning, with the help of the idea of FML, we propose an edge synergy-empowered FML model for few-shot malicious traffic detection against APT attacks. In our method, the malicious traffic detection model is distributed by the central server, and the edge-side nodes jointly maintain the central model using the

meta-learning method. At the same time, MAML algorithms maintain the personalization of the local model. In the model training phase, we deployed more than 400 edge devices to simulate real distributed scenarios. All edge devices will jointly maintain the central model. In the classification phase, we compared the five-classification capabilities of two different classification networks, FPN and Resnet, for malicious network traffic.

The main contributions of our work are as follows.

- 1) A privacy-preserving few-shot traffic detection (PFTD) framework against APT attacks is proposed based on federated meta learning. In the PFTD training phase, only 10 samples in each type of attack methods participate in the model updating process, maximizing the simulation of real scenarios with a limited number of APT attack samples. Besides, our proposed framework realizes the detection function for unknown samples with leaving data from local devices.
- 2) An edge synergy-empowered FML training algorithm is also designed to improve the accuracy of proposed PFTD framework. On each edge devices, the model-agnostic meta learning (MAML) unit can quickly obtain a personalized FML model that adapts to local small sample data through several steps of gradient descent. At the same time, the great generalization ability brought by the global model ensures that unknown samples can be identified, according to its poor prediction values in the classification results of each category.
- 3) To improve the adaptability of proposed PFTD to the heterogeneous threat intelligence samples collected by each edge device, we also propose a novel support-query mutual guidance mechanism to associate the support set and the query set, and then quickly achieve a local personalized model by fine-tuning.

The remainder of the article is organized as follows. Section II discusses the related work. Section III formulates challenges, motivations and system model, while Section IV describes our proposed FML method and training steps. Section V introduces the experiment. Section VII draws conclusions.

II. RELATED WORK

In this section, we will first introduce the abnormal traffic detection method for the APT attack, and then the research work related to Federated Meta Learning is discussed.

A. Anomaly Traffic Detection Against APT Attacks

Personalized anomaly traffic detection system is a personalized network traffic anomaly detection method proposed by us. Traditional network traffic anomaly detection methods often use machine learning and deep learning frameworks to detect abnormal network traffic. There are many machine learning methods applied to network traffic anomaly detection, such as random forest (RF), support vector machine (SVM), naive Bayesian network (NBN), logistic regression and other methods. Since traditional machine learning models only learn the attack patterns of simple features in TCP/IP packets. The models often exhibit high false positive rates [12]. Therefore, deep learning is

proposed, in which CNN can comprehensively learn the complex hierarchical feature representation of TCP/IP data packets; RNN can remember past information in a large number of TCP/IP packets, and the detection efficiency is greatly improved. Xiao et al. [13] proposed an intrusion detection model based on feature dimensionality reduction and convolutional neural network, which extracts and analyzes data features for classification by convolutional neural network. The experimental results show that the prediction accuracy of the model can reach 94.00%. Althubiti et al. [14] used a long short-term memory network, a special recurrent neural network (RNN), to build an intrusion detection model, which achieved a prediction accuracy of 84.83% on the CICDS001 data set. In addition, many methods have further improved the detection accuracy by combining the models. Andresini et al. [15] proposed an intrusion detection model that combines generative adversarial network (GAN) and convolutional neural network (CNN). The model achieves 93.29% accuracy on the KDD99 data set. Hassan et al. [16] building an intrusion detection model by combining a CNN with a long short-term memory network. The experimental results show that the prediction accuracy of the model can reach 97.17%.

In recent years, the design of edge-side intrusion detection system has become a new research direction. Kumar et al. [17] proposed the development of distributed intrusion detection systems (DIDS) using emerging and promising technologies such as blockchain on stable platforms in cloud infrastructure. The collaborative intrusion detection system (CIDS) developed by Sharma et al. [18] protects edge devices against cyber attacks such as denial of service (DoS) and distributed denial of service (DDoS) under the multi-access edge computing (MEC) architecture. After 2 years, Sharma et al. [19] proposed a new pure edge-based hybrid CIDS architecture based on CIDS to complement the previous work. Nie et al. [20] proposed a GAN-based multi-stage deep learning framework to achieve intrusion detection against multiple attacks at the edge network. Yin et al. [21] proposed a data collaboration mechanism that offloads the training model to distributed edge devices to achieve cooperative privacy protection for IoT.

There are many detection methods for APT attacks. In recent years, APT attack detection methods based on provenance graphs have achieved good results. The team at the University of Illinois at Chicago proposed the SLEUTH [22] method in 2017. Based on log data, it uses causal relationship tracking and provenance graphs to construct a model and finally reconstructs APT attacks. On this basis, the team introduced two methods, Poirot [9] and Holmes [7] in 2019. Poirot introduced cyber threat intelligence (CTI) and graph matching algorithm based on provenance graph. Holmes designed the high-level scenario graph (HSG) to realize the mapping of low-level (log, alarm) information to high-level. In 2021, the team further proposed the Extrator [23] method. It introduces natural language processing (NLP) into the CTI report, so as to realize the precise extraction of attack behavior in CTI. It also uses semantic role labeling (SRL) for semantic analysis to understand the relationship between attack behaviors and convert unstructured text into provenance graphs. In addition, due to the lack of explanation for decision-making in AI-based CTI and the massive deployment of edge devices, Li et al. [24] proposed an

explainable intelligence-driven APT edge defense mechanism. This approach increases the level of protection and defense against APTs at the edge.

B. Federated Meta Learning

Federated meta learning (FML) is the framework that combining federated learning and meta-learning [25]. Fallah et al. [26] used meta learning to implement personalized and differentiated training for federated learning. The federated learning (FL) framework is first proposed as standard federated training (FedAvg) [27], [28], [29]. FL has made great contributions in ensuring information security during Big Data exchange, protecting terminal data and personal data privacy. Wu et al. [30] designed a blockchain-empowered privacy preservation architecture for 5G-enabled drone communications. Yang et al. [31] present a design of blockchain-based crowdsourcing FL system for IoT devices manufacturers to learn customers better, the use of differential privacy (DP) protects the privacy of customer data and improves the accuracy of the model. Besides that, the blockchain will audit all client updates during federated training so that the system can be held accountable for model updates to prevent malicious clients or manufacturers. Moon et al. [32] reviewed the application of FL methods in the field of healthcare, including COVID-19, mammograms, and brain tumor segmentation. They believed that compared with traditional centralized models, FL-based models improved the model performance. While FL models in wearable healthcare do not significantly outperform centralized models, FL provides better results in terms of data privacy. Li et al. [33] proposed a federated deep learning method called DeepFed to develop a novel federated learning framework for multiple industrial CPSs that can collectively build a comprehensive intrusion detection model in a privacy-preserving manner. As the most representative method in few-shot learning, model-agnostic meta-learning (MAML) [34], which treats the designed meta-learner as an optimizer that learns to update the parameters of any model, was proposed to solve the problem of small amount of data and strong heterogeneity in edge devices in federated learning. The work of Jiang et al. [35] and Li et al. [36] corroborate that the FML model is a personalized global model with strong generalization ability. Apart from that, FML has made great contributions in multiple fields. A FML method proposed by Zhao et al. [37] reduced the BER performance and complexity of classical matched filter (MF)-based detectors; it demonstrated that FML is useful in an acoustic radio cooperative wireless network environment outperforms federated learning-based systems and has higher generalization ability. Zhang et al. [38] used a federated meta-learning approach to improve the accuracy of wireless traffic prediction at the edge. Additionally, FML has made significant contributions in several fields. This work trains an inductive network through meta-learning, applies the FSL architecture to NLP tasks in a federated environment, reduces the communication frequency between nodes, and has good compatibility for devices with small amounts of data [39]. Jiang et al. [40] proposed a Federated Meta-Location Learning (FMML) method that improves the accuracy of fine-grained location prediction on smartphones in 5 G scenarios. The FML model designed by Elgabli et al. [41] can be fine-tuned to new tasks with a small number of samples

in a distributed environment, and the model reduces the energy consumption of computation.

The success of federated learning depends on the fact that the data in each data center is independent and identically distributed (IID). Personalized federated learning (PFL) can solve the problem of data heterogeneity in federated learning and the lack of personalization of models for local tasks or data sets. Tan et al. [42] divided PFL into two categories: global model personalization and learning personalized model. Global model personalization focuses on the training ability of the global FL model on non-IID data, while learning personalized model focuses on achieving model personality during the training phase. Considering that this article mainly focuses on the problem of non-independent and identical distribution of edge device traffic data, we mainly research articles related to global model personalization. [43], [44], [45] These three papers all add a regular term on the basis of the federated average. The FedProx algorithm dynamically controls the number of updates of each node in each round of communication through the gamma inequality; the novelty of the L2GD algorithm is that it controls node training and server aggregation through probability; the advantage of the Ditto algorithm is that the local model is first optimized according to the FedAvg algorithm, and then a regular term is added to continue to optimize the local model, and finally optimization results are sent to the server without a regular term. Fallah et al. [26] regarded the training phase of FL as the training phase of meta-learning, regarded the personalization phase of FL model as the testing phase of meta-learning, and migrated the meta-learning model to personalized federated learning.

III. PROBLEM FORMULATION

A. Challenges and Motivation

Traditional malicious detection models are often deployed at the center of distributed systems. This tends to present some challenges:

- 1) Less fault tolerance, central models tend to be fragile when faced with multivariate and heterogeneous data. The retraining process of the model is accompanied by a huge waste of resources.
- 2) The volume of traffic data aggregated by distributed devices is huge, and the traffic data of distributed devices is unbalanced. Data cleaning, preprocessing, standardization, and normalization will lead to huge consumption of computing resources.
- 3) There is a delay in the response time of the malicious detection system deployed in the center, which is often caused by two reasons. On the one hand, it is the computing delay caused by the huge amount of data, and on the other hand, it is the physical delay caused by the distributed devices being far away from the central server.

The advantages of meta-learning itself can improve the efficiency of malicious detection models. First, meta-learning can deal with the problem of data imbalance brought by distributed systems. When faced with the problem of data imbalance, meta-learning can process the head and tail data differently, adaptively learn how to re-weight, or formulate a domain adaptive problem. Second, meta-learning has a strong generalization ability, it

performs few-shot learning by training a model on a set of source tasks that is only a few gradient descent steps away from a good task-specific model. Deploying meta-learning models on edge devices can meet the personalized learning needs of non-IID data, and the source tasks distributed by the central server can match the personalized task models of edge devices after several steps of gradient descent.

Therefore, building malicious detection models with federated meta-learning will bring many benefits. First, deploying the model on edge devices in a distributed environment can effectively alleviate the computational cost of the central server and greatly reduce the response time. Secondly, the introduction of meta-learning can help the central model to quickly adapt to local data samples, and the generalization ability is greatly enhanced; the model's processing ability to unbalanced data is greatly improved.

B. System Model

In traditional machine learning tasks, algorithms tend to focus on one specific task. In a network traffic-based attack behavior discovery system, a basic task is to use a classifier to label network traffic. That is, we have K samples and M labels denoted as follows: $\mathcal{D} = \{(x_1, y_1), (x_2, y_2), \dots, (x_i, y_j) \dots (x_K, y_M)\}$, $x_i \in \mathbb{R}$, $y_i \in \{0, 1, 2 \dots M\}$. The main goal is to build a multi-class classifier where x_i is the input to the model, y_j will be the output, and y_j is a possible estimate of the sample labels. In the traditional supervised learning scenario, the number of samples K is very large, and the data set containing all samples will be simply divided into two parts $\mathcal{D}_{(train)}$ and $\mathcal{D}_{(test)}$. However, in APT attack behavior detection, the number of samples is limited, and the entire task will be regarded as a small sample learning task. The conditional supervised learning method often used in few shot learning will face the problem of overfitting. In addition, in a distributed scenario, the traffic data samples collected by each device are not balanced, so the intrusion detection model should be retrained in order to adapt to new local data. Besides, the attack behavior detection system for APT often maps the attack behavior to the cyber kill chain. To address these differences, we propose the FML framework to implement abnormal traffic detection, and accordingly, as shown in Fig. 1, our proposed method no longer focuses on a single task but on multiple tasks, allowing the model to learn. The entire federated meta-learning model is divided into three levels. The bottom layer presents data sources which cover gateways, firewalls and base stations. In our model, each edge node represents an independent edge device, and each edge device will train a local personalized model through meta-learning on the basis of the initial model and update the parameters of the local model to the central server. The weight parameters will be aggregated at the server and new parameters counted by the aggregation algorithm will be redistributed to each device.

In the FML framework, considering the limited number of advanced persistent threats samples, the training task of the device will be regarded as a few-shot learning task. In addition, the introduction of the FML framework will solve the problem of unbalanced traffic data on edge devices. The specific method will be described in Sections 4.1 and 4.2.

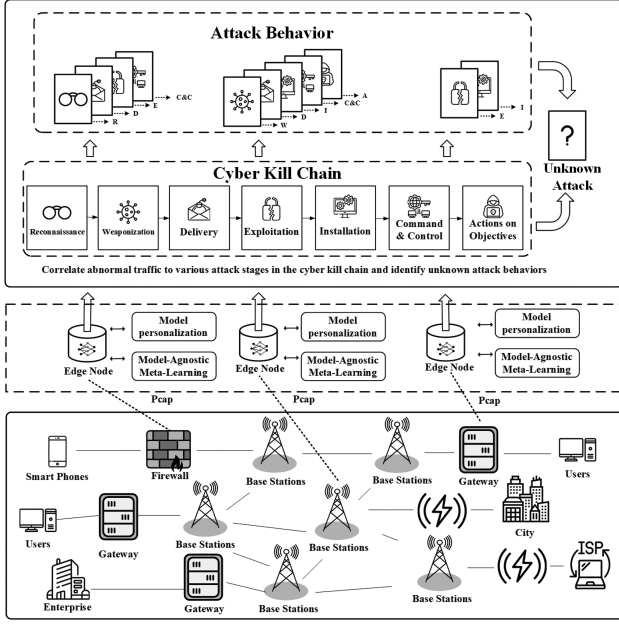


Fig. 1. An overview of Federated Meta Learning in staged APT detection at edge devices.

TABLE I
SYMBOL LIST

No.	Symbol	Explanation
1	O	Benign traffic
2	A to M	13 different types of attack samples
3	S_{train}	Training set
4	S_{test}	Testing set
5	K	Number of samples
6	S_u	A union of samples from A to I
7	$S'_u[i]$	Sampled from S_u
8	S_v	A union of samples from J to M
9	$S'_v[j]$	Sampled from S_v
10	θ_0	Initialization parameter
11	U_t	Union of edge devices
12	g_u	Loss Function
13	n	The number of tasks involved in training
14	α	The global learning rate
15	$F_i(\omega)$	The optimization objective
16	k	Communication rounds
17	β	The learning rate of local update

IV. THE PFTD WORKFLOW

In abnormal traffic detection scenarios, where the number of samples with supervised information is limited or impossible to obtain, few-shot learning is proposed to learn from a limited number of samples with supervised information. For the convenience of reading, the explanation of related symbols is shown in Table I.

A. Few-Shot Learning Formulation

In the FSL task, the dataset D is divided into two parts D_{train} and D_{test} . Training set $D_{train} = \{(x_i, y_i)\}_{i=1}^I$ where I is small, testing set $D_{test} = \{x^{test}\}$. In FSL we denote the ground truth joint probability distribution of input x and output y by $p(x, y)$. The $\hat{h}$ represents the optimal hypothesis from x to y , which is discovered by fitting D_{train} and testing on D_{test} . The FSL

Algorithm 1: FedMeta with MAML.

// Run on the server

Input: Initial iterate θ_0

Output: θ (Add-6)

Algorithm Update:

Initialize θ_0 for MAML.

for each episode $t = 1, 2, \dots$ **do**

Sample a set U_t of n edge devices, and distribute θ_0 (for MAML) to the sampled devices.

for each device $u \in U_t$ in parallel **do**

Get test loss $g_u \leftarrow \text{ModelTrainingMAML}(\theta_0)$

end

Update algorithm paramers $\theta \leftarrow \theta_0 - \frac{\beta}{m} \sum_{u \in U_t} g_u$ for MAML.

end

(M-1) // Run on edge devices u

ModelTrainingMAML (θ):

Sample support set S_{train} and query set S_{test}

$\mathcal{L}_{S_{train}}(\theta) \leftarrow \frac{1}{|S_{train}|} \sum_{(x,y) \in S_{train}} \ell(f_{\theta}(x), y)$

$\theta_u \leftarrow \theta - \alpha \nabla \mathcal{L}_{S_{train}}(\theta)$

$\mathcal{L}_{S_{test}}(\theta_u) \leftarrow \frac{1}{|S_{test}|} \sum_{(x',y') \in S_{test}} \ell(f_{\theta_u}(x'), y')$

$g_u \leftarrow \nabla_{\theta} \mathcal{L}_{S_{test}}(\theta_u)$

Return g_u to server

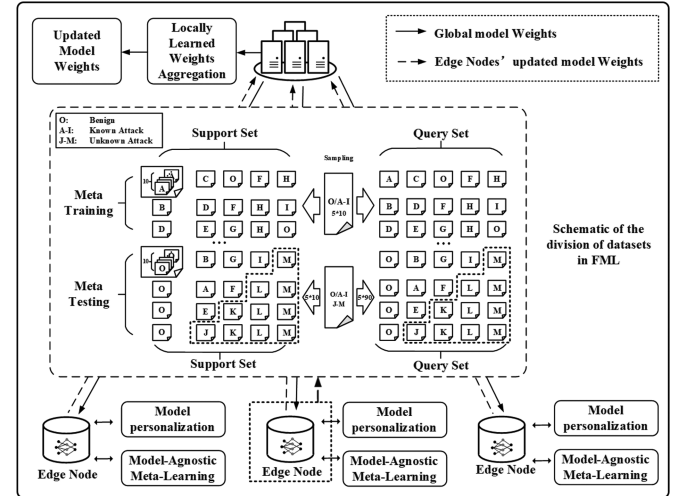


Fig. 2. The system model of proposed PFTD based on federated meta learning.

model defines a hypothesis space $\mathcal{H}$, which consists of $h(\cdot; \theta)$'s, to approximate the $\hat{h}$. FSL is a continuous optimization process whose purpose is to searches $\mathcal{H}$ to find the θ that parameterizes the best $h^* \in \mathcal{H}$. The performance of FSL is measured by the loss function $\ell(\hat{y}, y)$, where $\hat{y}$ is the predicted value and y is the observed output.

B. Edge Synergy-Empowered FML Model Training

The structure of the FML model is shown in Fig. 2. Suppose there are N devices and 13 different attack types of traffic data in the federated meta-learning scenario. The edge synergy-empowered feature of the FML model is reflected in the training

process. The entire FML model will complete the training process under the cooperation of these N edge devices. Each edge device will pass the parameters to the central server and get the initial parameters by means of average aggregation. In the meta-training stage, 400 devices (from 0 to 399) will be used to train a well-initialized model. In the meta-testing stage, 4 devices (from 400 to 403) will be used to verify the generalization ability of the model and the ability to identify unknown samples. The traffic data contains 14 categories, O represents benign traffic and from A to M are 13 different types of attack samples. In the meta-training stage, considering the limited number of APT attack samples, each device extracts 4 types of traffic data from $A - I$ with O as support and query sets S_{train} , each type has K samples. We define $S_u = \{A, B, C, D, E, F, G, H, I\}$ samples a set $S'_u[i]$ of i types of attack data from S_u (i from 0 to 4). $S_{train} = \{S'_u[4], O\}$.

In the meta-testing stage, in order to verify the generalization of the model against non-IID local data and the ability to identify unknown attacks, we trained J - M samples on the test device according to different proportions. We define $S_v = \{J, K, L, M\}$ samples a set $S'_v[j]$ of j types of attack data from S_j (j from 1 to 4). On each edge device we extract 4 types of attack data from A to M with normal traffic data O as support and query set S_{test} . $S_{test} = \{S'_v[j], S'_u[4 - j], O\}$.

The goal of traditional FL can be expressed as:

$$\min_{w \in \mathbb{R}^d} f(w) := \frac{1}{n} \sum_{i=1}^n f_i(w) \quad (1)$$

The optimization objective can be defined as:

$$f_i(w) := \mathbb{E}_{(x,y) \sim p_i} [l_i(w; x, y)] \quad (2)$$

The algorithmic structure of the entire federated meta-learning with MAML is shown in Algorithm 1, which modified from the research of Chen et al. [46]. The training process of MAML includes two levels of parameter update and transfer. First, the initialization parameter θ_0 is used to train the task-specific model, and then the loss of all tasks is used to update the initialization parameters of the model. Under the federated setting, each client downloads the initialization parameters of the model from the server, uses the support set to train the model, and then sends the test loss obtained by the query set to the server. The server side saves the initialization of the model and collects the test loss sent back by the clients of each mini batch. The messages transmitted between server and client are only model initialization parameters and test loss. Specifically, in Algorithm update stage, n edge devices sampled from U_t will receive an initial parameter θ for MAML process. After model training, loss function g_u is calculated and the new parameters θ is send to the server. In model training stage, each edge device samples support set S_{train} and query set S_{test} . During the training process, the support set and the query set guide each other and transfer knowledge. Different from the traditional FL method, the parameters finally passed by the edge device are generated under the mutual guidance of the support set and the query set. The whole process is shown in the Model Training MAML process in Alforithm 1. The loss function g_u will return to server.

Algorithm 2: Personalized FedAvg Algorithm.

Input: Initial iterate w_0 , fraction of active users r .

Output: T_{r_j}

for $k : 0$ **to** $K - 1$ **do**

Server chooses a subset of edge devices $\mathcal{A}_k$

uniformly at random and with size rn ;

Server sends w_k to all edge devices in $\mathcal{A}_k$;

for $i \in \mathcal{A}_k$ **do**

Set $w_{k+1}^i = w_k$;

for $t : 1$ **to** τ **do**

Compute the stochastic gradient

$\tilde{\nabla} f_i(w_{k+1,t-1}^i, \mathcal{D}_t^i)$ using dataset $\mathcal{D}_t^i$;

Set

$\tilde{w}_{k+1,t}^i = w_{k+1,t-1}^i - \alpha \tilde{\nabla} f_i(w_{k+1,t-1}^i, \mathcal{D}_t^i)$;

Set $\tilde{w}_{k+1,t}^i = w_{k+1,t-1}^i - \beta (I -$

$\alpha \tilde{\nabla}^2 f_i(w_{k+1,t-1}^i, \mathcal{D}_t^i)) \tilde{\nabla} f_i(w_{k+1,t}^i, \mathcal{D}_t^i)$;

end

Edge device i sends $w_{k+1,\tau}^i$ back to server;

end

Server updates its model by averaging over received

end

C. Personalizing to Individual Edge Devices

In order to generalize the model from a single edge device to multiple devices and ensure that each edge device can obtain a model adapted to local data with only a few steps gradient descents. We implement federated meta-learning using the Per-FedAvg method [26]. The algorithmic structure of the entire FedAvg method is shown in Algorithm 2. For each edge device, its meta function $F_i(\omega)$ can be defined as:

$$F_i(\omega) := f_i(\omega - \alpha \nabla f_i(\omega)). \quad (3)$$

The optimization objective can be defined as:

$$\min_{\omega \in \mathbb{R}^d} F_i(\omega) := \frac{1}{n} \sum_{i=1}^n f_i(\omega - \alpha \nabla f_i(\omega)). \quad (4)$$

Here α is the learning rate, n is the number of tasks involved in training. This formula could ensure that the central model can capture the differences between different edge devices, and each device can modify the initial model according to its own data. Algorithm 2 describes the entire FML implementation process. The optimization objective $F_i(\omega)$ is the average of all meta-functions. On each edge device, the gradient of the local functions $\nabla F_i(\omega)$ needs to be computed first:

$$\nabla F_i(\omega) = (I - \alpha \nabla^2 f_i(\omega)) \nabla f_i(\omega - \alpha \nabla f_i(\omega)). \quad (5)$$

Since computing $\nabla f_i(\omega)$ is expensive, so take a batch of data D^i to get an unbiased estimate:

$$\tilde{\nabla} f_i(\omega, \mathcal{D}^i) := \frac{1}{|\mathcal{D}^i|} \sum_{(x,y) \in \mathcal{D}^i} \nabla l_i(\omega; x, y). \quad (6)$$

At the k th round of communication, the server sends the current global model to several randomly selected edge devices, and each edge device performs t rounds of local gradient updates.

TABLE II
THE ATTACK TYPES OF CICIDS2017 AND DAPT2020

DAPT2020	CICIDS2017	
Benign	Benign	Web Attack Brute Force
Directory Bruteforce	FTP-Patator	Web Attack XSS
Network Scan	SSH-Patator	Web Attack Sql Injection
Web Vulnerability Scan	DoS Hulk	Infiltration
Account Discovery	DoS GoldenEye	Bot
Account Bruteforce	DoS slowloris	DDoS
SQL Injection	DoS Slowhttptest	PortScan
Backdoor	Heartbleed	

We optimize the meta-model based on the support set, where β is the learning rate of the local update, i represents user i , $k+1$ refers to using the meta-model parameters of the k th round for training, and t is the number of local iterations:

$$\omega_{k+1,t}^i = \omega_{k+1,t-1}^i - \beta \tilde{\nabla} F_i(\omega_{k+1,t-1}^i). \quad (7)$$

The last step is to use the query set to calculate the loss and its gradient to $\omega_{k+1,t}^i$ based on the optimized meta-model $\omega_{k+1,t}^i$:

$$\begin{aligned} \tilde{\nabla} F_i(\omega_{k+1,t-1}^i) &:= (I - \alpha \tilde{\nabla}^2 f_i(\omega_{k+1,t-1}^i, \mathcal{D}_t^i)) \\ &\tilde{\nabla} f_i(\omega_{k+1,t-1}^i - \alpha \tilde{\nabla} f_i(\omega_{k+1,t-1}^i, \mathcal{D}_t^i), \mathcal{D}_t^i). \end{aligned} \quad (8)$$

Each edge device sends the local $\omega_{k+1,t}^i$ to the server, and the server updates the global model by calculating the average of ω_{k+1} :

$$\omega_{k+1} = \frac{1}{r\eta} \sum_{i \in \mathcal{A}_k} \omega_{k+1,t}^i \quad (9)$$

V. EXPERIMENTS

A. Datasets

Our proposed method is evaluated on the DAPT 2020 dataset. Only csv data obtained from CICFlowMeter feature extraction was used. External and internal network traffic were mixed together. The original data table contains 85 columns with 83 features and 2 labels. Columns 1 to 7 of the table data are traffic identification features, and all of them except the protocol column were removed. Another 12 columns were removed because they were recorded the same on all traffic. The CICIDS2017 dataset contains benign and recent common attacks similar to real-world data (PCAPs).

The DAPT2020 dataset contains 5 days of network traffic, including normal traffic and 7 kinds of APT attack traffic. CICIDS2017 contains 5 days of network traffic data, including normal traffic and 14 types of malicious traffic. The specific attack types of the two datasets are shown in Table II. As shown in Fig. 2, in the training phase, we will randomly sample 4 types of malicious traffic, plus normal traffic, a total of five types of traffic data. For each type of traffic, we will draw 10 samples as data sources to build tasks. Each task is defined as a few-shot five-classification task to distinguish between normal traffic and different kinds of malicious traffic. In the testing phase, we select 4 types of traffic as unknown traffic to test the detection accuracy of the FML model when there are unknown types of attacks locally. Specially, he classification task on the

TABLE III
MULTI-CLASSIFICATION ACCURACY CONFUSION MATRIX

Accuracy		Prediction				
		A	B	C	D	E
Actual	A	TP1	FP1	FP2	FP3	FP4
	B	FP5	TP	FP	FP	FP
	C	FP6	FP	TP	FP	FP
	D	FP7	FP	FP	TP	FP
	E	FP8	FP	FP	FP	TP

DAPT2020 dataset will eventually map data traffic to the five stages of APT attacks.

B. Metrics

Accuracy and macro F1 score are used as evaluation metrics. Accuracy represents the ratio of correctly classified samples to all samples. When the data is unbalanced, the accuracy rate cannot reflect the classification effect of the class with a small number of samples. In addition to the accuracy index, the macro average F1 score is used as the evaluation criterion, which is the average of five F1 scores as shown below.

$$Macro F1 = \frac{\sum_i^m F1 score_i}{m} \quad (10)$$

The basic unit of meta-learning is a task, so the evaluation of meta-learning performance needs to calculate the accuracy of classification results according to the confusion matrix, as shown in Table III. For multivariate classification tasks, the classification results correspond to the following two results:

- 1) True Positive (TP): Normal samples are classified as normal;
- 2) False positive (FP): Normal samples are classified as malicious (false positive);

Based on this, precision can be defined as shown below.

$$Accuracy = \frac{TP1}{TP1 + FP5 + FP6 + FP7 + FP8} \quad (11)$$

C. Experimental Settings

Experiments were performed using the following hardware and software platforms: Intel(R) Xeon(R) Platinum CPU @ 2.50 GHz, 128 GB RAM, NVIDIA GeForce RTX 3090, Ubuntu 20.04.3 LTS, CUDA 11.3, and PyTorch 1.12.1.

To apply the federated meta-learning approach, the dataset is resampled and distributed across different edge devices. The dataset is divided into 70% and 30% parts for training and testing edge devices respectively. In particular, the 15 traffic data in the data-stealing stage are manually divided into the latter. For training and testing edge devices, we set up their classification tasks in different ways. For training edge devices, we split the data by “activity”. Each edge device randomly has data of 5 activities, and each activity has 15 traffic. For the test edge device, we split the data by “stages”. Each edge device has 15 flows per attack phase. Finally, each edge device has a different 5-classification task to simulate the different distribution of traffic classes on each node. For all edge devices, 10 samples of each class are used as the support set and 5 samples are used as the query set.

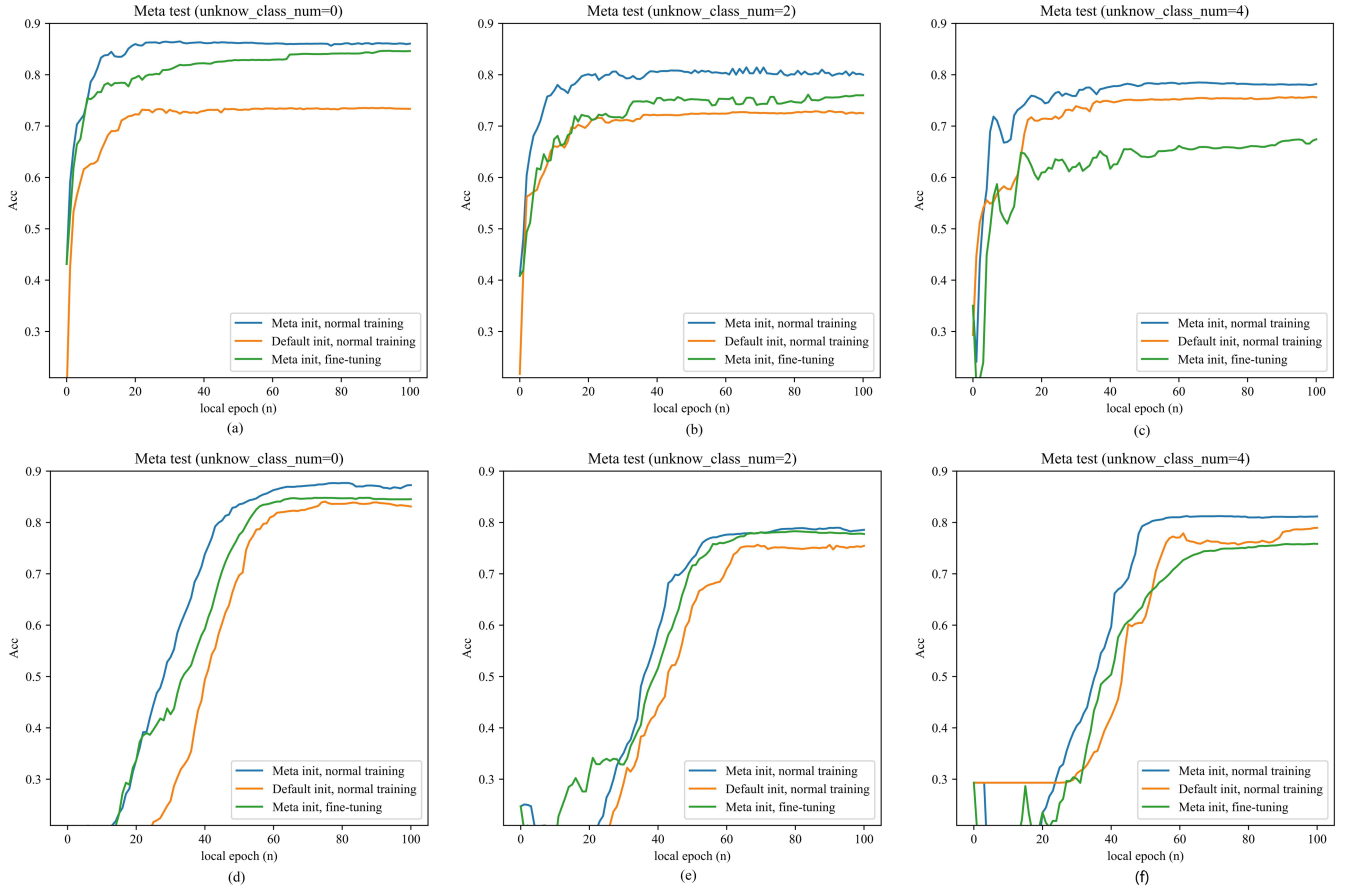


Fig. 3. Detection results of experiments on the CICIDS2017 dataset.

D. Results

1) *Compatibility Test*: In the FML scenario, the number of edge devices affects the training effect of the model. We compared the impact of different numbers of edge devices on the accuracy of the model when the local unknown sample types are both 2. The experimental results are shown in Fig. 4. Two conclusions can be drawn:

- 1) The more edge devices, the higher detection rate of the model. The increase in the number of edge devices allows the central model to fit a greater number and variety of samples. Of course, this improvement is not unlimited. When the number of edge devices is sufficient to cover enough sample types, the accuracy of the model will no longer improve.
- 2) Scenarios with a small number of edge devices are not suitable for deploying FML models. When the number of edge devices is only 20, the model accuracy is lower than default initialization. The limited number of edge clients means that they vary greatly, which is not guaranteed that the central model has learned enough sample features during the training process. This leads to experimental results lower than default initialization. Meanwhile, When the number of edge clients increases by a small amount, the models learned by the central server are quite different. When the client base is large enough, the number of clients will increase less and has less impact on the overall model

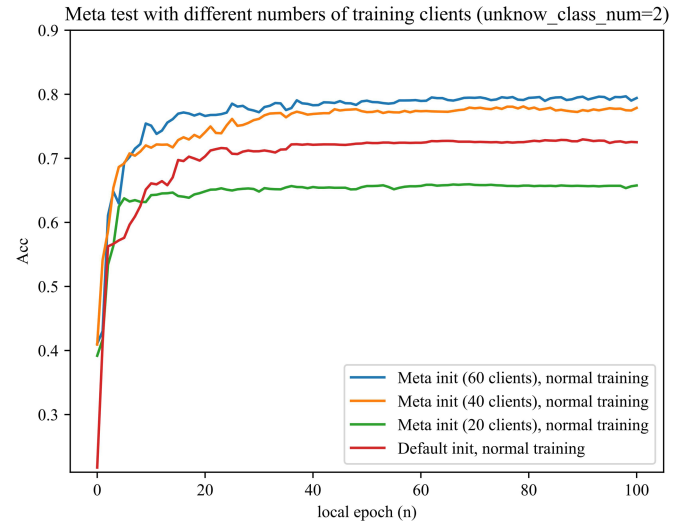


Fig. 4. Comparisons of the detection accuracy against different numbers of edge devices.

variance. This leads to the fact that the offset between 20 and 40 is larger than that of 40 and 60.

2) *Comparative Experiment*: We tested our method on two different datasets: CICIDS2017 and DAPT2020. The detection results of experiments on the CICIDS2017 dataset are shown in

TABLE IV
COMPARISON OF DETECTION RESULTS, THE NUMBER OF SAMPLES AND CLASSIFICATION METHOD IN THE PROPOSED METHOD AND RELATED RESEARCH WORKS

Method	Dataset	Feature Type	Classification Method	Number of Samples	ACC (%)
RFA(2018)	ISCX2012	Statistic+bigram encoded payload	Two classifications	25	77.60
FC-Net	ISCX2012FS	CNN	Two classifications	5	97.56
FC-Net	ISCX2012FS	CNN	Two classifications	10	97.87
FC-Net	CICIDS2017FS	CNN	Two classifications	5	94.33
FC-Net	CICIDS2017FS	CNN	Two classifications	5	94.64
Proposed FML	CICIDS2017	FCN	Multiple classifications	10	86.07
Proposed FML	CICIDS2017	ResNet	Multiple classifications	10	87.27
Proposed FML	DAPT2020	FCN	Multiple classifications	10	69.20
Proposed FML	DAPT2020	ResNet	Multiple classifications	10	67.60

Fig. 3. In this experiment, we compare the accuracy of anomalous traffic detection systems in three scenarios, training the samples in central server, deploying a federated meta-learning model, and fine-tuning a local model before testing. The neural networks used for classifying are FCN (Fully Connected Network) and Resnet. Form Fig. 3(a) to (c) and from Fig. 3(d) to (f), the number of unknown class in attack data was set differently. The experimental results show that, comparing with training the samples in center server, deploying the federated meta-learning model can improve the edge device's recognition accuracy of attack samples. In the case of a small number of unknown attack types, the detection accuracy of the test edge device after model fine-tuning is also improved compared with the accuracy of the traditional central server training model. However, the results of model fine-tuning drop when the number of unknown attack types grows. This result occurs because the local sample is quite different from the sample fitted by the central model.

The detection results of experiments on the DAPT2020 dataset are shown in Fig. 5. This experiment studies a five-category problem, and malicious traffic will be mapped to the five attack stages in the APT attack. The settings of this experiment remain unchanged, and the neural networks used for classification are FCN and Resnet. We compared the detection accuracy on default initialization, federated meta learning and FML with fine-tuning. In this experiment we did not test the model for unknown class samples recognition ability, but instead focuses on detecting the detection ability of FML in different datasets. The results show that the application of FML improves the detection accuracy of traditional centralized training while using FCN as the classifier. The difference from the results of the CICIDS2017 dataset is that the accuracy produced by model fine-tuning is not stable above the default initialization. The reason for this phenomenon is that different attack stages often have traffic data generated by similar system behaviors. The FCN network does not fully learn this classification method. However, when resnet is used as the classifier, the results of the three cases do not show a large difference as the number of training rounds increases. The occurrence of this phenomenon has a certain relationship with the residual structure of the resnet model. The deeper network layer makes the classification results of the model relatively stable, so under the effect of FML and fine-tuning, the degree of discrimination is not obvious.

Detailed results of comparative experiments are shown in Table IV. In the test of the CICIDS2017 dataset, we add different

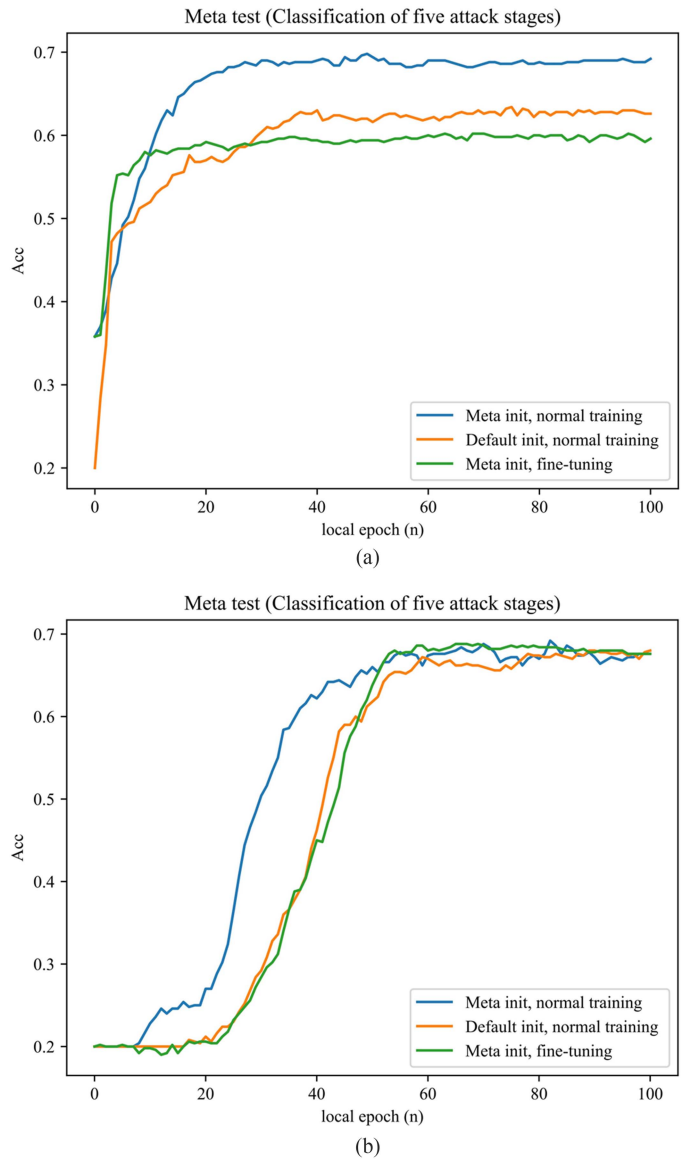


Fig. 5. Detection results of experiments on the DAPT2020 dataset.

types of unknown samples. The results show that under a FCN classification network, when the number of unknown species is small, the application of FML can improve the detection

TABLE V
COMPARISON OF DATASETS, UNKNOWN ATTACK NUMBER AND DETECTION RESULTS IN PROPOSED METHOD

Dataset	Feature Type	Unknown Attack Number	Default Init		Meta Init			
			Normal Training		Fine-tuning		Normal Training	
			ACC	F1	ACC	F1	ACC	F1
CICIDS2017	FCN	<i>Unknown_num</i> = 0	0.73	0.69	0.85	0.84	0.86	0.86
		<i>Unknown_num</i> = 1	0.71	0.64	0.80	0.75	0.83	0.80
		<i>Unknown_num</i> = 2	0.73	0.60	0.76	0.66	0.80	0.75
		<i>Unknown_num</i> = 3	0.74	0.61	0.72	0.66	0.83	0.79
		<i>Unknown_num</i> = 4	0.76	0.66	0.67	0.62	0.78	0.73
	ResNet	<i>Unknown_num</i> = 0	0.83	0.83	0.85	0.84	0.87	0.87
		<i>Unknown_num</i> = 1	0.81	0.78	0.82	0.79	0.85	0.82
		<i>Unknown_num</i> = 2	0.75	0.70	0.78	0.72	0.79	0.73
		<i>Unknown_num</i> = 3	0.80	0.74	0.79	0.72	0.84	0.78
		<i>Unknown_num</i> = 4	0.79	0.72	0.76	0.68	0.81	0.75
DAPT2020	FCN	Five Stage Classification	0.63	0.57	0.60	0.52	0.69	0.68
	ResNet	Five Stage Classification	0.68	0.67	0.68	0.66	0.68	0.67

accuracy by about 10%, and finally reach more than 80%. Model fine-tuning also has better detection accuracy. When the number of unknown species is large, the accuracy rate decreases. Due to the powerful classification function of ResNet, the FML does not improve the accuracy rate significantly, but it also has a certain effect. The first reason for the difference in accuracy improvement is that different DNN networks have different matching degrees with data. Resnet may be more suitable for classifying traffic data, so the improvement in accuracy rate is not as obvious as FCN. Secondly, another reason is that Resnet's residual structure has more advantages than FCN in the face of deep networks. Resnet solves the problem of gradient disappearance encountered in the case of deepening the network. So in the default training stage, it has a higher accuracy rate. In the test of the DAPT2020 dataset, we classified the five APT attack stages corresponding to the samples. The experimental results show that under the FCN classification network, the application of FML improves the accuracy of the model, and the F1 score also improves. However, due to the classification performance of the Resnet network, in the FML scenario, the classification accuracy has not improved.

Considering that the classification networks in CICIDS2017 and DAPT2020 are both five-category, we conducted a migration experiment of the model, which trained the model on CICIDS2017 and tested it on DAPT2020. In the FCN network, as shown in Fig. 6(a), after 40 rounds of local edge device training on the DAPT2020 dataset, the central model learned from CICIDS2017 can well adapt to local samples, and the accuracy rate increases to 70%. The accuracy of the model after fine-tuning is several percentage points higher than the previous experiment. In the resnet network, as shown in Fig. 6(b), after 40 rounds of training, the accuracy of the model is the same as the previous experiment. It can be seen that the FML model has certain migration capabilities.

Table V shows the comparison of detection results, the number of samples and classification method in the proposed method and related research works, which is extended from the work of Xu et al. [47]. On the whole, there is a certain gap between our detection accuracy and previous work, but considering that we use a multi-classification network, and FML will have a certain

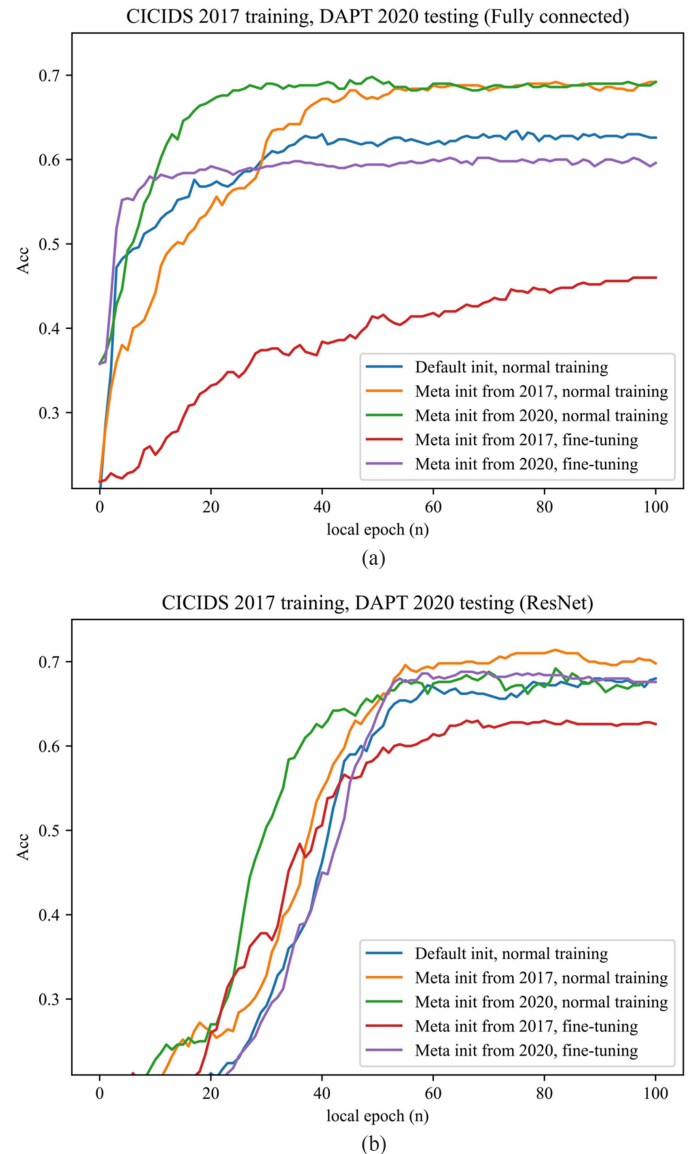


Fig. 6. Detection results of experiments when training on CICIDS2017 and testing on DAPT2020.

impact on the results in the selection of model aggregation algorithms, so consider from this perspective, our proposed method is feasible.

VI. CONCLUSION

In this article, we proposed a novel detection method based on federated meta learning framework to implement few-shot abnormal traffic detection. For this purpose, a virtual federated meta-learning scenario with hundreds of edge devices and two different five-class classification networks: FCN and ResNet were constructed for learning. The proposed method accomplishes this task by training a FML-based abnormal traffic training and detection model. In order to adapt to the characteristics of APT attacks, we only use a small number of labeled samples for training, and use FCN and ResNet to implement five-category tasks and identify unknown attack behaviors. On the edge device, we use the MAML method to realize the personalized adjustment of the central model to the local data, so that the central model can quickly adapt to the local data and achieve high classification accuracy. At the same time, the combination of MAML and federated learning protects the information security and privacy of edge devices. Experimental results show that the proposed small-shot APT detection method achieves state-of-the-art performance in FML scenarios. Our work shows that federated meta-learning methods can maintain the performance of APT attack detection methods in distributed scenarios, which will inspire other researchers to design better deep neural networks for APT attack detection along this direction.

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